

Machine Learning Insights Into Eating Disorder Twitter Communities

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ABSTRACT

Eating disorders have serious impacts on young population physical, psychological, and social functioning. Health agencies are calling for new psycho-therapeutic intervention considerations that take into account online communities, which is not possible if therapists know little about eating disorders discussions online. In this paper, we leverage machine learning analytics to understand what the eating disorder communities are talking about over time and how they are talking about it. By analyzing local and global community discussions, we discovered complex group dynamics underpinning collective identities offering emotional support but also potentially perpetuating harmful behaviors. Our analysis of four local subcommunities and four global theme evolutions revealed prevalent subjects, perspectives, motivations, and linguistic patterns. We found tight-knit communities grounded in shared membership, goals, cultures, and practices. Community voices highlighting recovery journeys were limited. Our computational assessment of invisible online spaces aims to inform personalized interventions accounting for community forces in youth mental health.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

KEYWORDS

Eating Disorders, Social Media, Discussion Analysis

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1 INTRODUCTION

The 2023 CDC report issued stark warning about youth mental health. Between 2011 and 2021, the number of teenage girls who felt sad or hopeless continued to rise sharply, from 36% to 57%, while among teenage boys they also increased from 21% to 28% during this 10-year period [6]. Furthermore, 30 percent of teenage girls seriously considered suicide, compared with 14 percent of teenage boys. The White House announced executive actions to protect youth mental health, safety, and privacy online [31]. The Surgeon General urged immediate action to ensure social media environments are healthy, safe, and transparent in response to the growing national youth mental health crisis [34]. The Congress held public hearings on social media and youth mental health, denouncing social media companies for their inaction on issues such as mental health, youth suicide, hate speech, online harassment, and eating disorders [35].

From 2019 to 2022, pediatric emergency department visits for eating disorders doubled among adolescent girls and increased by a quarter among adolescent boys [5]. People with eating disorders experience severe and persistent disturbed eating behaviors, accompanied by distressing thoughts and emotions [2]. Their disordered eating behaviors can lead to malnutrition that affect physical, psychological and social functioning [18]. Although there are different types of eating disorders, they have overlapping symptoms, including physical damage to cardiovascular, gastrointestinal, and musculoskeletal systems, psychological problems related to anxiety, depression, and fear, as well as behavioral issues such as social withdrawal, obsessive control, and self-harm [20]. Adolescents with eating disorders are at least three times more likely to have suicidal thoughts and two times more likely to attempt suicide [28]. Eating disorders are more common among adolescents, especially adolescent girls [21].

In 2023, Pew Research found that 93% of teenagers use social media [24]. They spend an average of 4.8 hours a day on social media [11]. These platforms are emotional refuge for many teenagers, with 67% of them saying they received heartfelt support on social media during difficult times [23]. The eating disorder community is active on many platforms [30]. Anyone with or without eating disorders can join the online communities and stay connected without any barriers. Eating disorder communities are not all about recovery, as many promote eating disorders as beneficial life choices rather than dangerous health conditions [9]. People who participate in

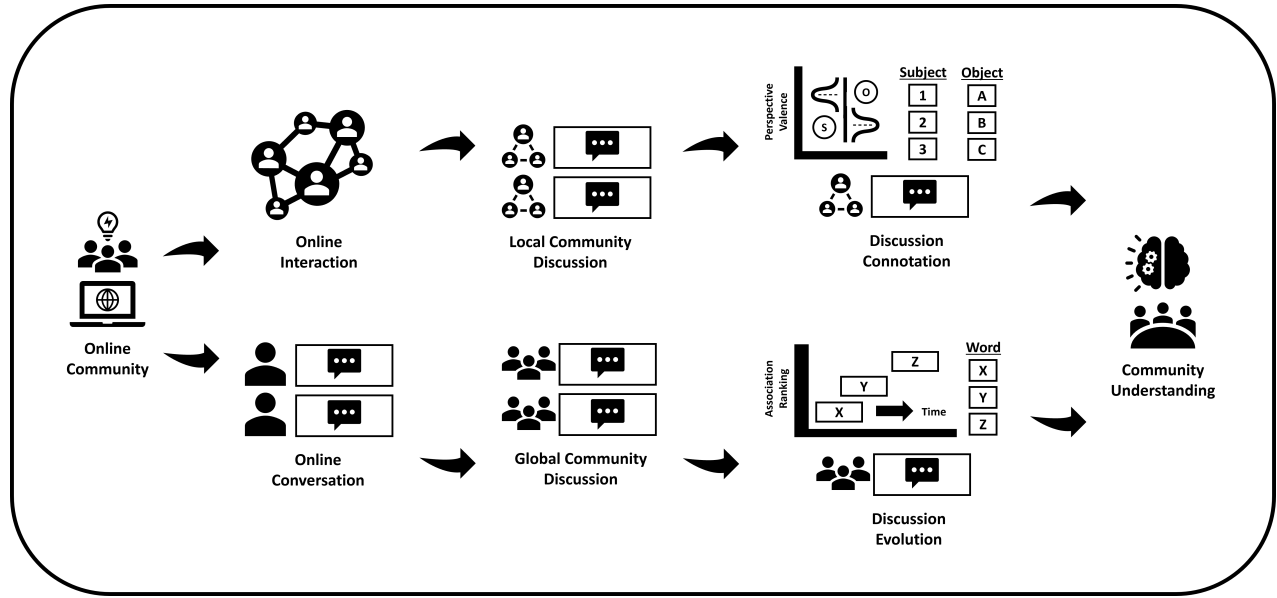


Figure 1: Analysis flowchart of eating disorder online communities. We focus on extracting local within community discussions from online interactions to perform connotation frame analysis. We perform global community discussion analysis using static embedding clustering and dynamic embedding similarity across communities.

such communities may develop, perpetuate, or exacerbate existing eating disorders. They are unknowingly trapped in a supportive environment filled with positive framing of eating disorders, sugar-coated harmful encouragement, and constructive-sounding body control [10]. Social media has become a risk catalyst for eating disorders, especially targeting youth as vulnerable populations.

Thus, it is crucial to re-calibrate psycho-therapeutic interventions for people with eating disorders need to account for online communities. However, therapists know little about eating disorders in online communities other than what their clients describe, and even the personal experiences shared by clients may not be the most accurate, complete, and truthful representations. This additional integrated knowledge will play a key role in psycho-therapeutic decision-making, planning, and treatment. This additional integrated knowledge will play a key role in psycho-therapeutic decision-making, planning, and treatment. In this paper, we leverage machine learning analytics to understand local and global eating disorder community discussions.

Main contributions:

- Our analysis revealed prevalent subjects, perspectives, motivations, and linguistic patterns.
- The findings showed complex group dynamics underpin collective identities grounded in shared membership, goals, cultures, and practices.
- We found the journey to recovery is not part of the larger discussion.

Machine learning is an important tool for quickly assimilating social media posts at scale and helping therapists to understand the online context, culture, and communication of eating disorders. Integrating such assessments with clinical expertise can strengthen

outreach and outcomes by framing technology as an ally rather than adversary.

2 RELATED WORK

Psychotherapeutic interventions follow decrease social media usage, deactivate social media, and decouple with online communities to address psychological issues related to social media. A randomized controlled trial found that significant improvements in depression severity among college undergraduates by limiting social media use to 30 minutes daily [13]. Another randomized controlled trial discovered that deactivation of social media for four weeks improved subjective well-being, self-reported happiness, and life satisfaction among young adults [1]. A longitudinal cohort study found that adolescents who use social media more than 3 hours per day at a higher risk for mental health problems [27]. Another randomized controlled trial discovered that taking a one week break on social media leads to significant improvements in well-being, depression, and anxiety [15]. Psychotherapeutic interventions often limit social media use. When today's generation is inseparable from social media, it is important understand online discussions.

Machine learning is widely used to detect eating disorders-related social media posts and online users with eating disorders. A model used domain-specific vocabulary, topics, psychological processes, and linguistic information of written posts to classify users with anorexia nervosa [25]. A RoBERTa model was trained to differentiate tweets written by people who suffered or not suffered from an eating disorder [4]. A multimodal model was used to classify if a pro-eating disorder content violates community guidelines based on textual and visual characteristics [7]. A XGBoost model was used to predict mental illness using social media posts and relevant comments [14]. A deep learning model was trained to learn linguistic



Figure 2: The perspectives of Recovery, EDTwt, Keto, and Ozempic community towards entities, top 10 subjects and objects. The perspective valence ranges from -1 to 1, where 1 indicates strong positive connotation.

markers of psychological disorders and language features to distinguish users diagnosed with a mental disorder and healthy users [33]. Topic analysis is then applied to understand eating disorder discussions by extracting eating disorder topics labeled with few representative words from classified eating disorder posts [16, 36]. Discussion analysis is oversimplified to what is being discussed

Unlike any other work, we collected eating disorder posts from Twitter based on relevant keywords provided by domain experts and leveraged machine learning analytics to understand the evolution of social media discussions within and across eating disorder communities. We specifically focus on discussion content ("what is being discussed within and outside communities over time") and community perspectives ("how it is being discussed")

3 DATASET

There are insufficient ways to identify people with eating disorders online without self-declaration. Therefore, we turned to studying eating disorders-related social media posts that certainly reflect some level of eating disorder online communities. We identified a list of keywords from prior works related to social media, eating disorders, and online discussions of eating disorders [8, 22]. Domain experts revised the compiled list, resulting in a final list of 48 keywords. We queried the Twitter API using these keywords with a maximum number of tweets of 5 million. Our dataset contain 2,630,049 eating disorders-related tweets from 557,099 Twitter users between October 2022 and March 2023. The collection period was chosen arbitrarily. We did not specify how many tweets to collect for each month. The monthly sample breakdown is 122,616, 763,718, 480,872, 523,855, 551,215, 187,773 tweets. From the data collected we

construct an eating disorder retweet network consisting of 43,832 Twitter users and 82,267 tweet interactions.

4 METHOD

We studied eating disorder online communities through local community discussion, global community discussion, discussion connotation, and discussion evolution shown in Figure 1. Local community discussions provide micro insights into eating disorder online subcommunities, while global community discussion provide macro insights into overall eating disorder online communities. These two layers of discussion analysis constitute a better understanding of eating disorder online communities.

4.1 Local Community Discussion

The local community discussion involved with smaller communities. We started with the retweet network for eating disorders and applied Louvain community detection [32]. The algorithm extracted a list of retweeting communities. The community list was filtered by outlier detection based on the number of Twitter users. In other words, the retweet community with few Twitter users were removed. Each retweet community had a word cloud and BERTopic lists generated from all its tweets [12]. We manually labeled retweet communities by comparing high-frequency words across the word clouds and BERTopic lists. A connotation frame were constructed for each retweet community. Connotation frames are a lexicon of transitive verbs that scores the perspective valence of associated subjects and objects [26]. The connotation considers the perspectives/subjective roles and relationships implied in the language instead of the overall perspective/valence of the text. This

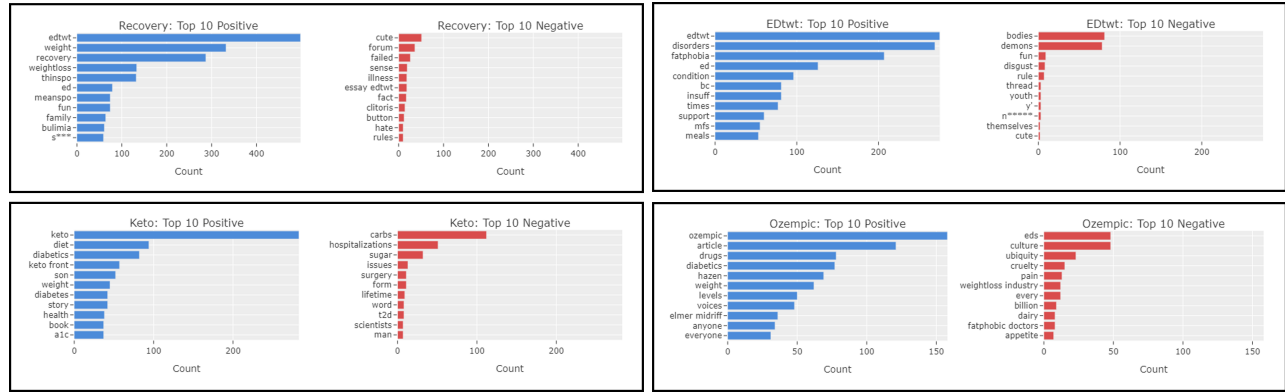


Figure 3: The top 10 positive and negative perspectives of Recovery, EDtwT, Keto, and Ozempic community towards entities.

gave us a detailed reading of perspective toward associated subjects and objects. The perspective valence ranges from -1 to 1, where 1 indicates strong positive perspective.

4.2 Global Community Discussion

The global community discussion involved entire communities. We started with tweets from all communities. The tweets were pre-processed, segmented into individual sentences, and broken down into individual tokens. Each token was labeled with a lemma, part-of-speech tag, and syntactic dependency. Some tokens were discarded based on part-of-speech tags and syntactic dependencies if they are additive to the overall semantics. The leftover tokens were merged to form subject phrase, verb, and object phrase tuples. A text was generated from each tuple. We embedded all text to capture sentence semantics [29]. The feature matrix was standardized and reduced with principal component analysis to retain 99% of the explained variance [17]. The agglomerative clustering algorithm extracted the number of clusters based on silhouette scores [19]. These clusters were denoised and partition into smaller clusters by comparing silhouette scores. We manually labeled the clusters from top 100 tweets in each cluster. ESTEEM visualized changes in related word for each cluster over time. ESTEEM is a spatiotemporal embedding visualization tool [3].

5 RESULTS

We analyzed what the retweet communities and the community as a whole were talking about on Twitter between October 2022 and March 2023. Our findings spanned local and global community discussions. There were overlap in what can be explained since global community discussions do include some degree of local community discussions. Local community discussion analysis focused on user perspectives, while global community discussion analysis focused on contextual changes.

5.1 Local Community Discussion Analysis

The local community discussion consisted of four retweet communities. They talked about recovery, EDtwT, keto, and Ozempic shown in Figure 2 and Figure 3. EDtwT is refers to the eating disorder community on Twitter. Ozempic is a drug used to treat type 2

diabetes, but is often used for weight loss. The recovery community often mentioned family as the subject and weight as the object in the sentence. They expressed positive perspective about recovery, weight, and family and negative sentiment about failed, illness, and hate. The EDtwT community often mentioned fatphobia as the subject and eating disorders as the object in the sentence. They expressed positive sentiment about EDtwT, fatphobia, and support and negative sentiment about body, disgust, and themselves. The keto community often mentioned keto as the subject and carbs as the object in the sentence. They expressed positive sentiment about diet, weight, and keto and negative sentiment about carbs, sugar, and scientist. The Ozempic community often mentioned article as the subject and Ozempic as the object in the sentence. They expressed positive sentiment about Ozempic, diabetic, and weight and negative sentiment about pain, fatphobic doctors, and appetite.

5.2 Global Community Discussion Analysis

The global community discussions included weight loss, EDtwT, keto, and Thinspo shown in Figure 4. Thinspo is short for thin inspiration. The keto, weight loss, and EDtwT talks went on for six months, while Thinspo talks happened on and off for only three months. There was large overlapped between EDtwT and Thinspo talks. There were meaningful insights from examining the temporal evolution of these discussions shown in Figure 5. Keto contents spanned kombucha, kelp, and herbal. Weight loss contents were consistent, focusing on overweight, obesity, and dieting. EDtwT contents included other online communities Anatwt, EDtwTThread, and Goretwt. Anatwt is refer to the anorexia community on Twitter, EDtwTThread is the eating disorder community on Twitter Thread, and Goretwt is the gore community on Twitter. There was interactions between the eating disorder community and other communities. Thinspo contents were consistent on fatspo and thin, but spanned allspo, nicespo, and anasp. These are different inspirations for body image. The eating disorder community talked about keto, weight loss, and EDtwT differently in focus and other discussions. Keto was discussed along with ingredients, kombucha, and kelp in focus discussions but other discussions were about Tiktok. EDtwT was discussed along with different online communities in focus discussions than other discussions. Weight loss was discussed along with overweight, obesity, and dieting in focus discussions but

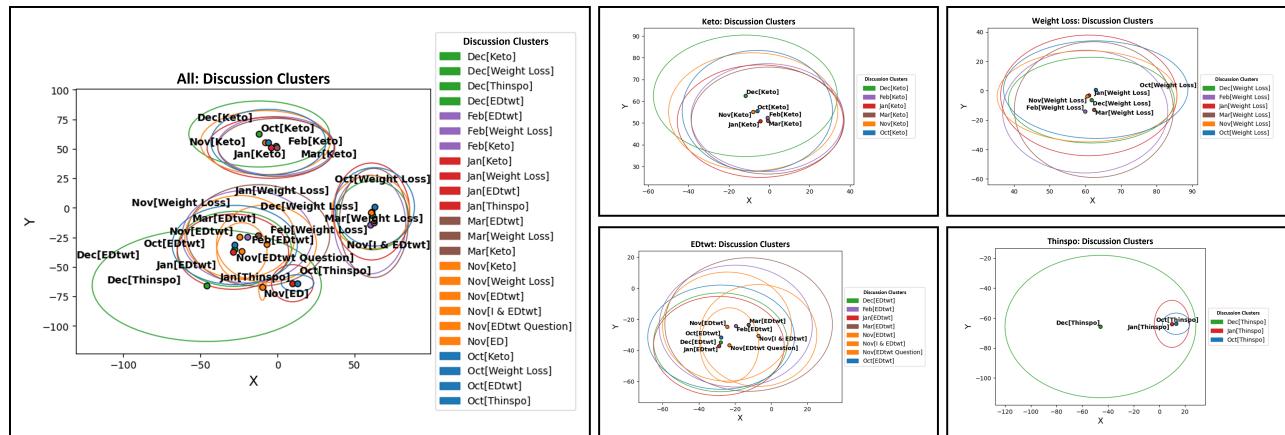


Figure 4: Monthly Twitter discussion content clustering in the embedding space across (left) and within (right) communities - Keto, Wights Loss, EDTwitter.

other discussions were about fat and pound. Thinspo was relatively the same between focus discussions and other discussions.

6 KEY FINDINGS

Understanding eating disorder online communities can help provide better guidance for people with eating disorders. Psychotherapeutic intervention emphasizes the importance of client listening, understanding, and feeling to a positive therapist-client relationship. These standards of practice are difficult to extend beyond individual clients to online communities due to cognitive limitations, information overload, and time constraints. To address this problem, we demonstrated how machine learning can help therapists to navigate large-scale, fast-paced social online environments and effectively listen, understand, and feel people with underlying health psychological issues in the online communities. These additional community understandings are critical to recalibrate psychotherapeutic interventions to account for online communities. The same psychological issues are more complex in social online environments than in real life. This layer of complexity remains to be explored.

We valued different voices in eating disorder online communities through analysis of local and global community discussions. The global community discussion covered weight loss, EDtwit, keto, and Thinspo. The eating disorder community expressed a sense of belonging from their conversations. EDtwit is a group membership when they often mention that their, their friend, people they know are involved in it. Thinspo is a culture of the eating disorder community who believe that thinness is an inspiration and they and others should work towards this goal. Weight loss is a common goal for people with eating disorders. The ketogenic diet is a popular option that caters to the interest of eating disorder community by restricting dietary intake, eating frequency, and weight gain. The shared membership, culture, goal, and practice come together to form a community identity that fosters personal connections, mutual understandings, and collective experiences. The eating disorder community is not just an online group, a collective identity.

The local community discussion covered recovery, EdTwt, keto, and Ozempic. The recovery discussion was a small voice within the eating disorder community. It is worth the effort to consider how to expand recovery discussions throughout the community. How the recovery group talked and felt about family, illness, and hate suggests that those factors are the driving force in eating disorder recovery. This provides a guidance for how therapists how can engage in recovery efforts within the community. The Edtwt group had a sense of belonging, strong community support, and a fat phobia. They become defensive against anyone criticizing their body, image, and attitude. The keto group believed this is the best diet for weight loss, and they rejected carbs, sugar, and scientists. The scientific claims against the ketogenic diet and weight loss are unconvincing to this group of people. The Ozempic group believed it is effective for weight loss and refused to listen to doctors about side effects. How they feel is just as important as what they say.

The local and global community discussions overlapped when it comes to EDtwit and keto but not when it comes to recovery, Ozempic, and Thinspo. The comparison of local and global community discussion showed the scale of a discussion. Ozempic is so popular for weight loss that it has appealed some people starting to talk about it, but it has not spread to the entire eating disorder community yet. Ozempic has the potential to become very popular among the eating disorder community in the future and caution is warranted if Ozempic appears in global community discussions. The Thinspo discussion was scattered among different people, rather than a group of people talking about it. That is why Thinspo appeared on and off in the global community discussion but not in the local community discussion. The EDtwit and keto discussions continued for six months, reflecting strong interest in local and global community discussions. Our analysis of local and global community discussions provided a deeper understandings of eating disorder online communities, which is helpful knowledge for therapists.

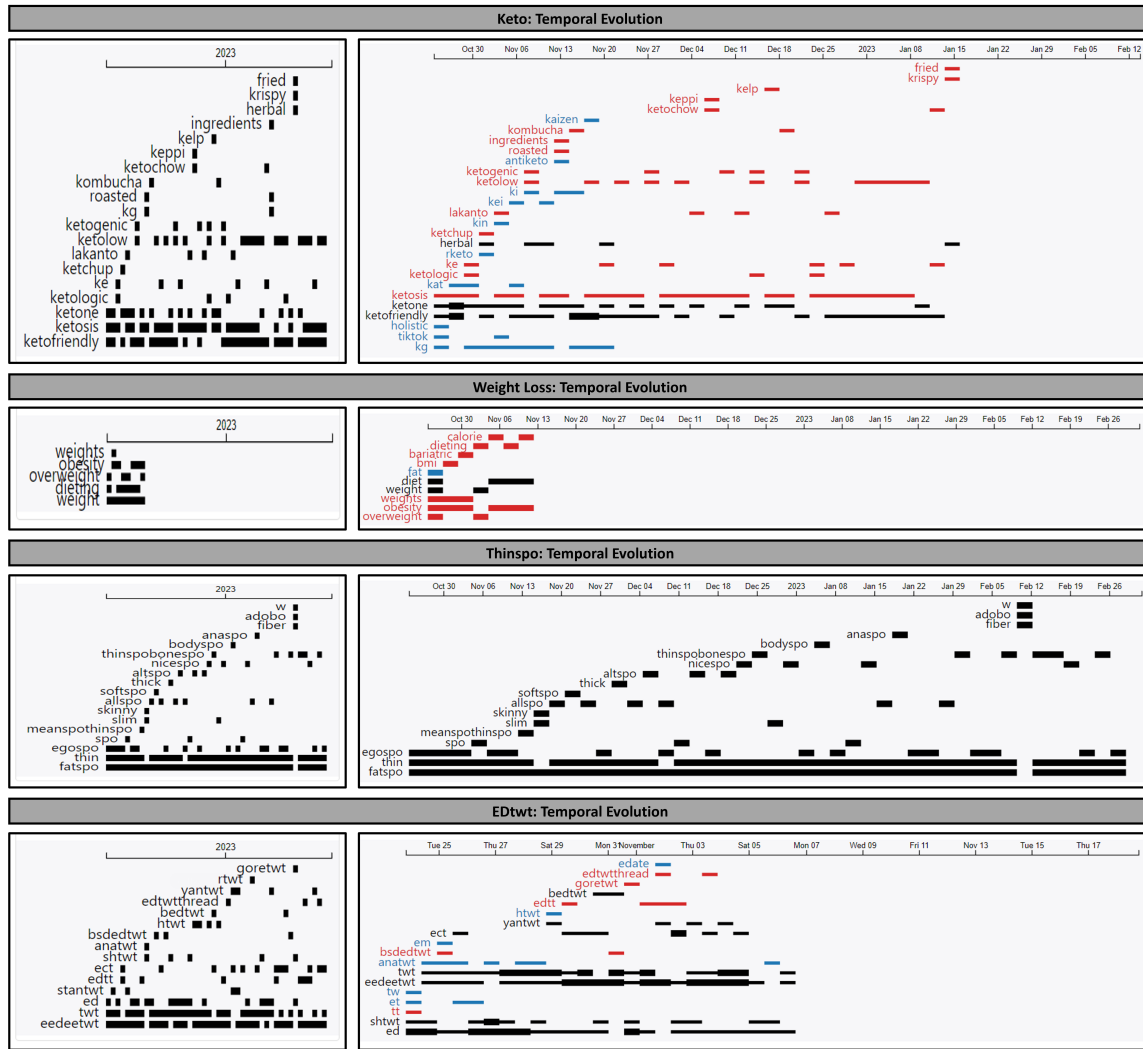


Figure 5: Eating disorder discussion evolution over time within each community. Rows represent content presence and similarity over time (left). Contrasting inside (blue), outside (red) and shared (black) community discussion evolution (right).

7 CONCLUSION

This study leveraged natural language processing and machine learning techniques to map online conversations in eating disorder communities on Twitter. By extracting both local subcommunity discussions and global theme evolutions, we gained a multilayered perspective into what eating disorder groups are talking about. Our computational analysis revealed complex dynamics underpinning collective identities in social media interactions and online communities. Beyond individual experiences, the broader community discourse plays an influential role for vulnerable teenagers – providing emotional support but also potentially perpetuating harmful behaviors. Critically, these online spaces remain largely invisible to practitioners developing one-on-one treatment plans.

While social media can exacerbate disorders, it also presents opportunities for support if properly understood. By illuminating key

subjects, perspectives, motivations, and linguistic nuances prevalent in eating disorder forums, this study aimed to bridge gaps between personal recovery and collective enablement. The frameworks and indicators extracted enable tailoring interventions to leverage or counterbalance community forces accordingly. This paves the way for personalized and scalable platforms extending care beyond clinical settings into digital environments where vulnerable youth already congregate. Extending analyses to additional domains and incorporating explanatory user models would further enrich collective insights to drive impact. Overall, integrating computational assessments of online ecosystems with qualitative therapeutic practices can strengthen outreach and outcomes for historically overlooked groups. Advanced analytics should complement clinical expertise, reframing technology as an ally rather than adversary in promoting healthy development.

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DATA BIAS

Twitter is an open social media platform that moderates certain eating disorder content. However, not all biased posts can be identified as targeting the eating disorder community. The dataset inevitably inherits some degree of individual or group bias toward eating disorder communities, practices, and conditions. We did not eliminate any bias toward eating disorder communities because it will not accurately reflect online discussions of eating disorders. It is critical to understand how bias plays into eating disorder discussions.

LIMITATIONS

This study has several limitations worth noting. First, our dataset comprised eating disorder-related tweets rather than tweets strictly from users with clinically verified conditions. The Twitter query relied on a predefined keyword list which shaped tweet sampling,

while monthly tweet volume fluctuations may further bias temporal analyses. Additionally, non-standard language, misspellings, abbreviations, and irregular grammar common in social media posts posed text processing challenges. From an analytics perspective, the algorithms used for community detection and discussion clustering impact extracted local/global topics. Other relevant but smaller communities and themes may be overlooked based on popularity signals or similarity metrics. Lastly, while this study focused specifically on textual content, multimodal machine learning incorporating images, videos, and networks could enrich insights.

In summary, our methodology makes best efforts at computationally assessing eating disorder perspectives from Twitter discourse. However, inherent data biases, textual ambiguities, analysis assumptions, and representation limitations temper the results. We aim primarily to spotlight online community forces strongly influencing youth mental health rather than derive absolute, generalized characterizations. Extending this work across multiple platforms and integrating explanatory user models would further strengthen reliability and precision.

ETHICS STATEMENT

This paper analyzed public Twitter data to study online eating disorder communities. We recognize the sensitive nature of this topic and have taken steps to conduct the research ethically. Our

analysis is based solely on the textual content of public tweets related to eating disorders. We did not attempt to diagnose or make judgments about any individuals. The perspectives extracted reflect discussion patterns in the data, not definitive conclusions about eating disorder sufferers. Moreover, the analysis seeks to highlight community dynamics rather than target any groups or individuals.

We cannot verify if Twitter users included in our dataset have clinically diagnosed conditions. Our findings characterize the online conversations in this dataset, not the eating disorder population at large. Significant variations may exist across platforms, demographics and personal experiences. We strongly discourage drawing generalized assumptions about people with eating disorders from this computational study alone. Qualified mental health experts should interpret the results critically, as Twitter data carries inherent biases.

Our intention is to demonstrate how advanced analytics can uncover hidden insights about vulnerable online groups. These additional signals aim to augment professional understanding of social forces shaping youth mental health. Integrating such assessments responsibly within therapeutic frameworks can aid personalization. We welcome dialog with researchers across disciplines to advance rigorous, thoughtful approaches analyzing complex health-related communities through emerging data sources. Collaborative efforts respecting a diversity of perspectives will provide deeper wisdom to support those suffering.